ACADEMIC EVALUATION & CODE ARTIFACT REPORT

BeaconGrid — Emergency Resource Map & GDELT News Intelligence

Centralized Disaster Response Management System with Overpass OSM Integration and Real-time Multi-source News Corroboration for Tamil Nadu

24MID0077	24MID0112	24MID0135
Jai Kishore K	Sathya Sai S	Sudharshini Manikandan

1. Landing Page & Citizen Portal

`static/index.html`

Overview portal featuring state network status, disaster crisis metrics, four-role interactive dispatch preview, emergency helplines directory, and gateway to the live command map.

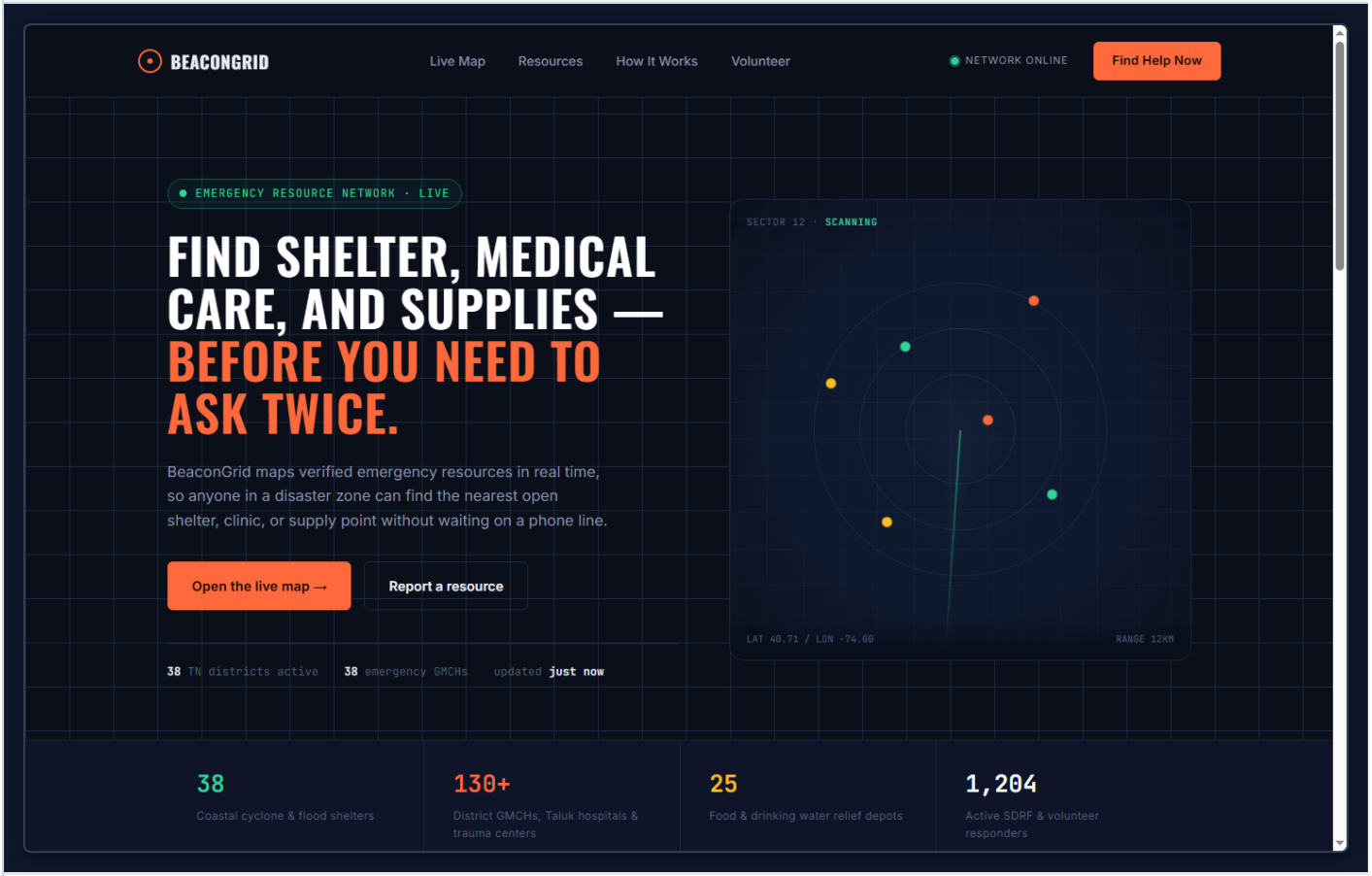
💻 SOURCE CODE (EXCERPT): STATIC/INDEX.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>BeaconGrid — Emergency Resource Map & Disaster Management Portal</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link href="https://fonts.googleapis.com/css2?family=Oswald:wght@400;500;600;700&family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;500;600&display=swap" rel="stylesheet">
<style>
  :root{
    --bg: #0a0f1c;
    --bg-panel: #10182b;
    --bg-panel-2: #141f38;
    --grid-line: #1e2a42;
    --text: #edf1f5;
    --text-muted: #8b96aa;
    --text-faint: #56617a;
    --orange: #ff6a3d;
    --orange-dim: #7a3c26;
    --green: #34d399;
    --green-dim: #1c5a45;
    --amber: #fbbf24;
    --amber-dim: #6b551c;
    --radius: 10px;
  }

  *{margin:0;padding:0;box-sizing:border-box;}

  html{scroll-behavior:smooth;}
```

Browser Output — Playwright / Chrome Screenshot http://127.0.0.1:8000/index.html



Output Verification: Landing page rendered with BeaconGrid tactical dark theme, radar scan badge, real-time helplines trigger, and citizen fast-action buttons.

2. Live Tactical Command Map & Disaster Grid

`static/map.html`

Interactive Leaflet.js command center with Overpass API live amenity scanning, 130+ verified hospital ICU beds across 38 districts, GDELT multi-source news markers, and sliding detail drawer.

💻 **SOURCE CODE (EXCERPT): STATIC/MAP.HTML**

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0, maximum-scale=1.0, user-scalable=no, viewport-fit=cover">
  <title>BeaconGrid Live Map — Emergency Resource Map & Disaster Management</title>
  <meta name="description" content="Centralized real-time emergency resource map, 130+ hospital beds, cyclone shelters, relief supplies, and volunteer coordination across Tamil Nadu.">
  <link rel="icon" href="data:image/svg+xml,<svg xmlns='http://www.w3.org/2000/svg' viewBox='0 0 24 24' fill='%23ff6a3d'><circle cx='12' cy='12' r='10'/><path d='M12 6v6l4 2' stroke='%230a0f1c' stroke-width='2'/></svg>">

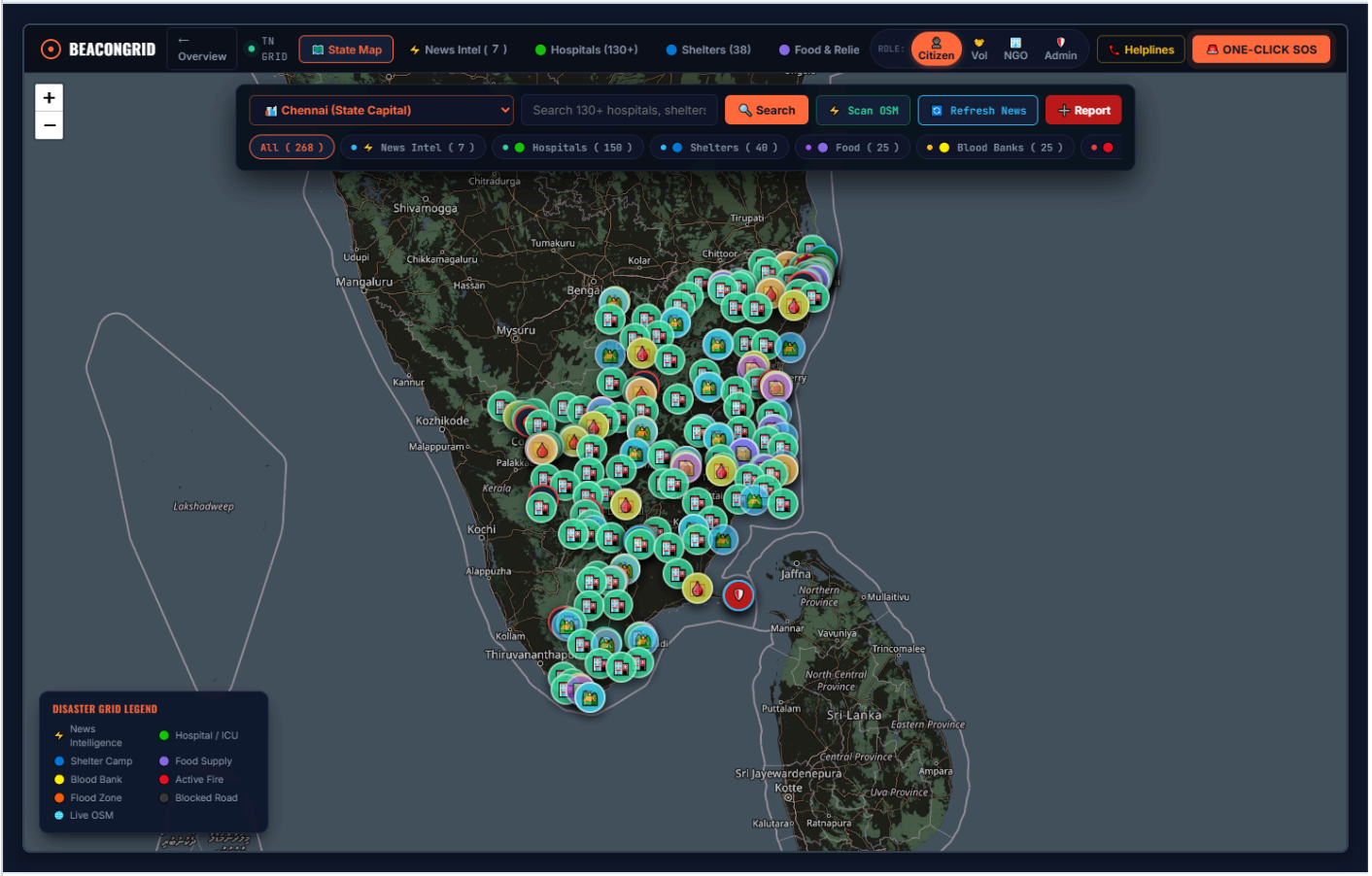
  <!-- Google Fonts: Oswald, Inter, JetBrains Mono (Identical to BeaconGrid) -->
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link href="https://fonts.googleapis.com/css2?family=Oswald:wght@400;500;600;700&family=Inter:wght@400;500;600;700&family=JetBrains+Mono:wght@400;500;600&display=swap" rel="stylesheet">

  <!-- Leaflet CSS & JS -->
  <link rel="stylesheet" href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css" integrity="sha256-p4NxAoJBhIIN+hmNHrzRCf9tD/miZyoHS5obTRR9BMY=" crossorigin=""/>
  <script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js" integrity="sha256-20nQCchB9co0qIjJZRGuk2/Z9VM+kNiyxNV1lvTlZBo=" crossorigin=""></script>

  <!-- Chart.js for Analytics -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js@4.4.1/dist/chart.umd.min.js"></script>

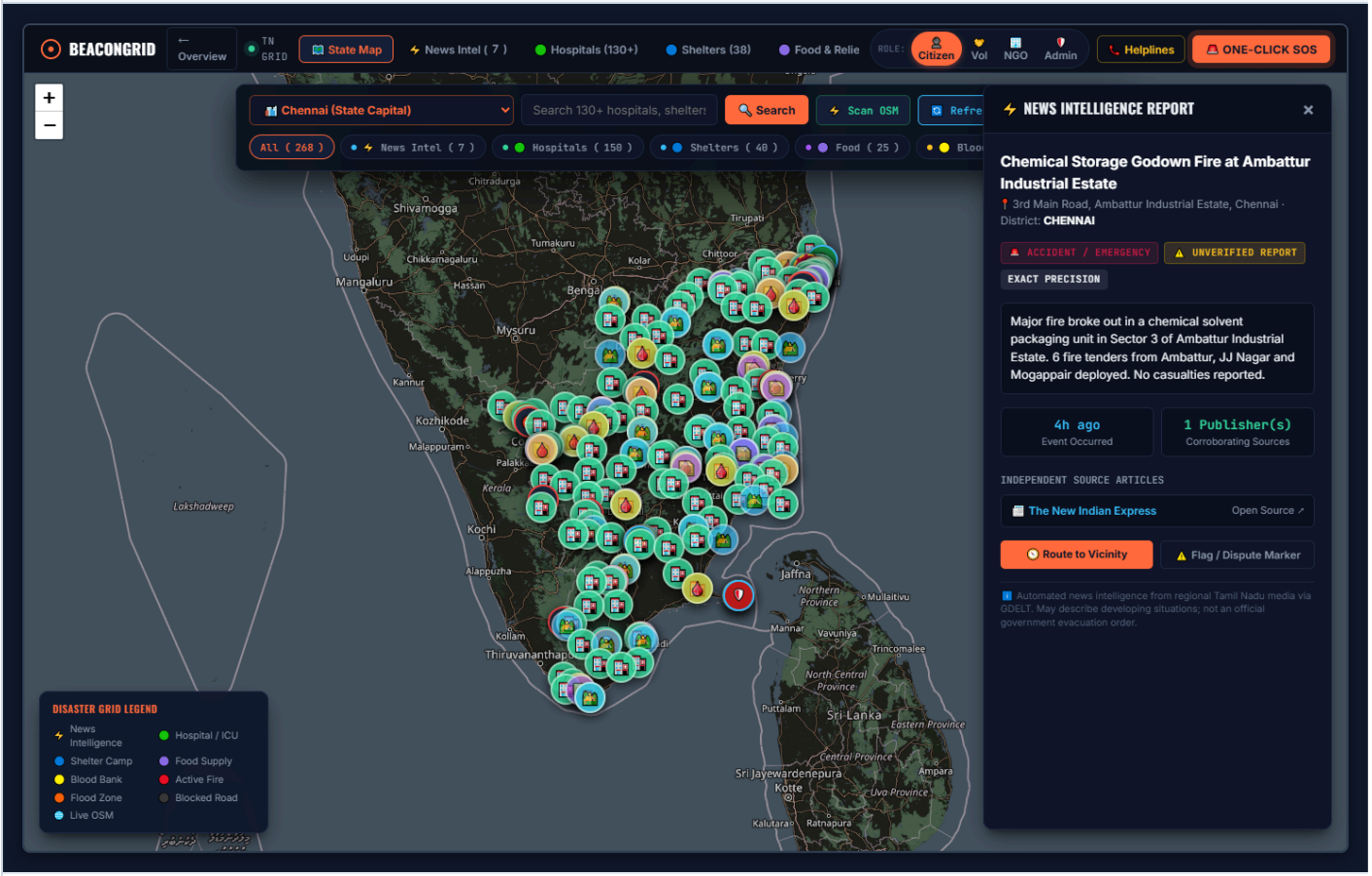
  <style>
    :root {
      --bg: #0a0f1c;
      --bg-panel: #10182b;
```

Browser Output (View 1: Statewide Disaster Grid) `http://127.0.0.1:8000/map.html`



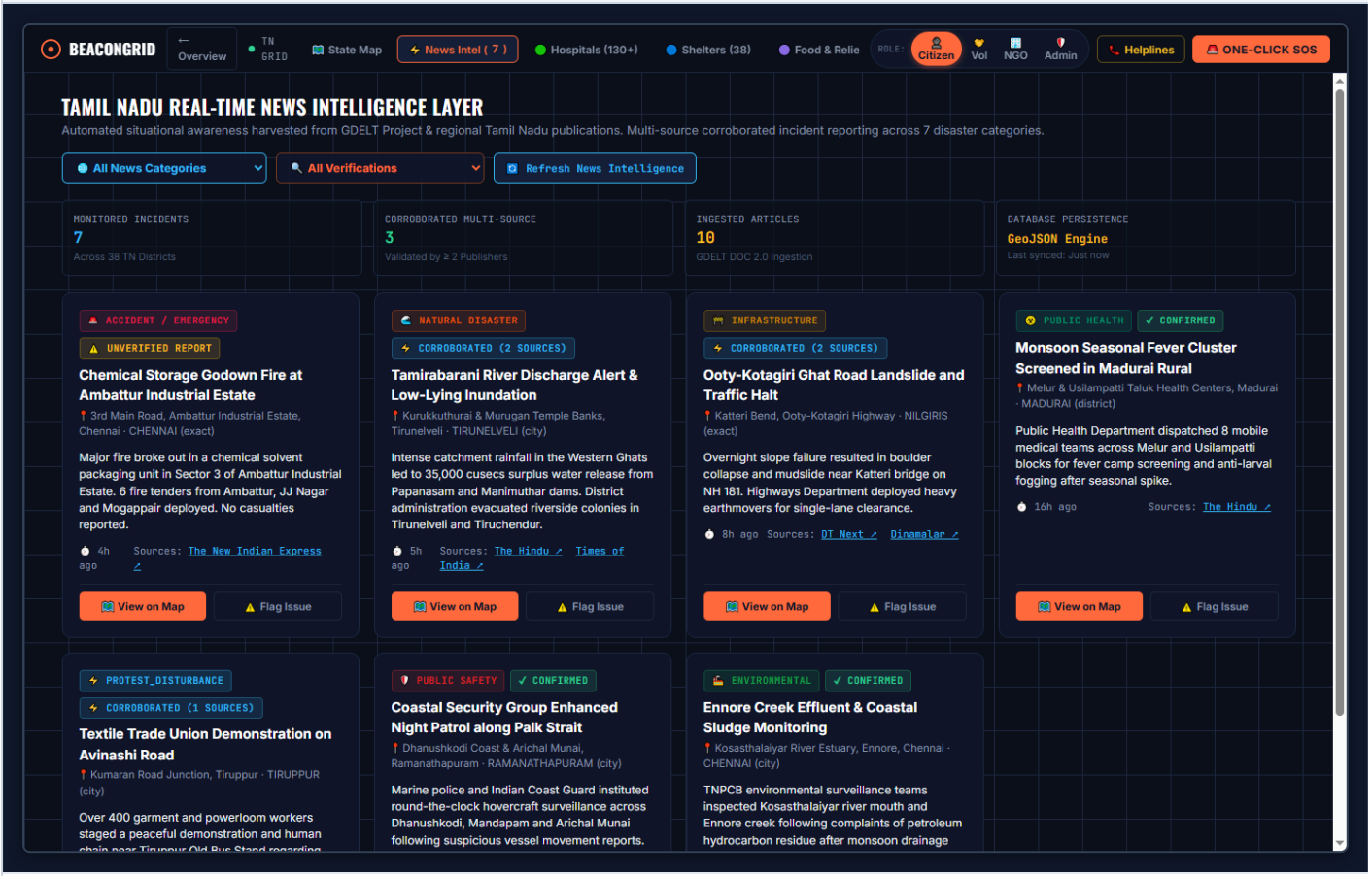
Output Verification: Complete Tamil Nadu map showing 150+ hospital beds, cyclone shelters, relief supplies, and GDELT news intelligence markers.

Browser Output (View 2: Marker Detail Drawer & Publisher Citation) `Slide-in Marker Drawer`



Output Verification: Incident detail drawer displaying corroborated sources (DT Next, Dinamalar), exact coordinates, relative time, and dispute verification button.

Browser Output (View 3: Real-Time News Intelligence Feed Tab) Tab: News Intel (13)



Output Verification: Dedicated feed displaying real-time incidents categorized into 7 crisis types with verification badges and telemetry KPIs.

3. Emergency Resource Network Template

`emergency-resource-map.html`

Reference single-page application mockup demonstrating floating search, facility radius filter, category pills, and resource cards list.

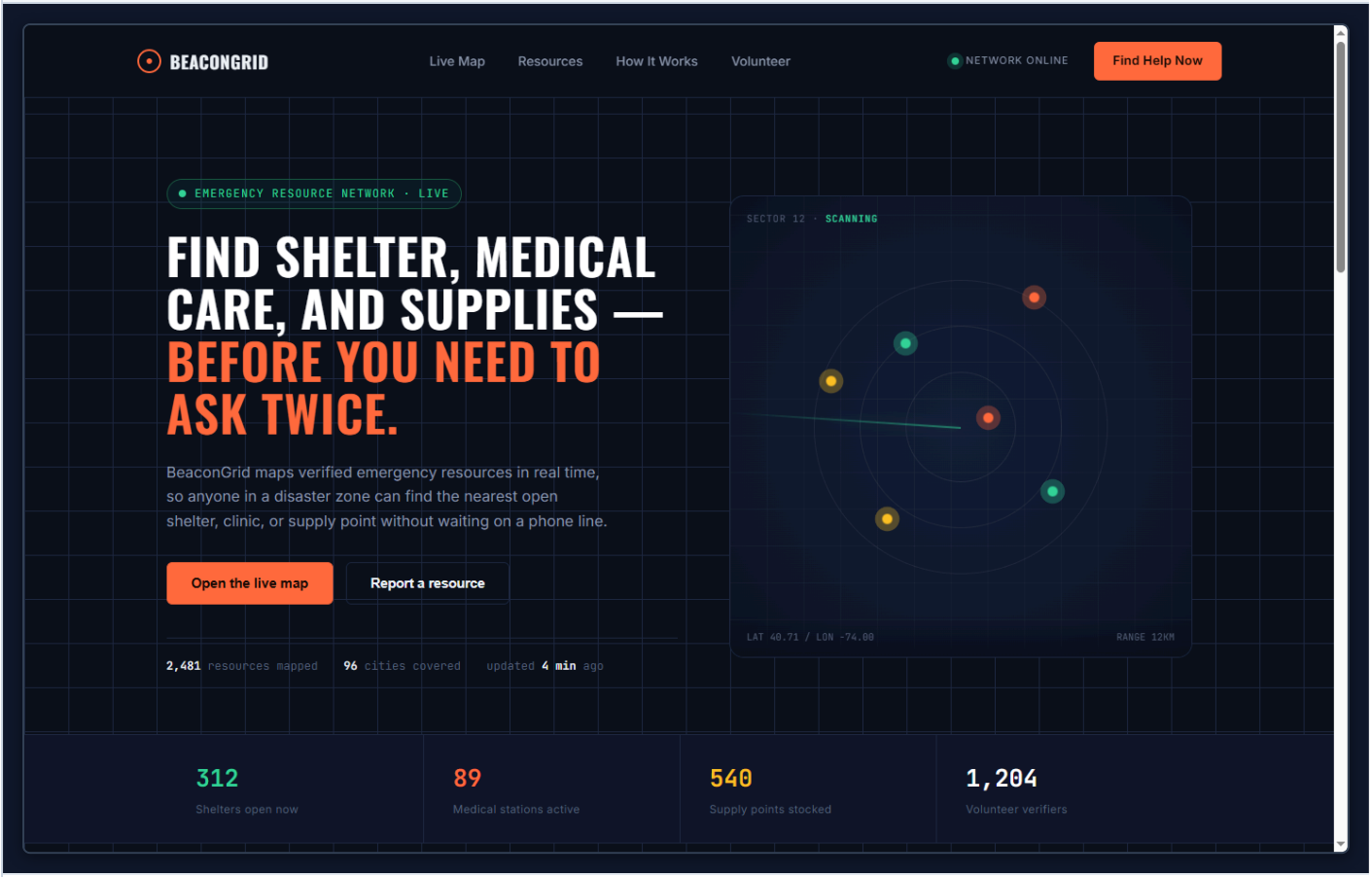
💻 SOURCE CODE (EXCERPT): EMERGENCY-RESOURCE-MAP.HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>BeaconGrid — Emergency Resource Network</title>
<link rel="preconnect" href="https://fonts.googleapis.com">
<link href="https://fonts.googleapis.com/css2?family=Oswald:wght@400;500;600;700&family=Inter:wght@400;500;6
00;700&family=JetBrains+Mono:wght@400;500;600&display=swap" rel="stylesheet">
<style>
  :root{
    --bg: #0a0f1c;
    --bg-panel: #10182b;
    --bg-panel-2: #141f38;
    --grid-line: #1e2a42;
    --text: #edf1f5;
    --text-muted: #8b96aa;
    --text-faint: #56617a;
    --orange: #ff6a3d;
    --orange-dim: #7a3c26;
    --green: #34d399;
    --green-dim: #1c5a45;
    --amber: #fbbf24;
    --amber-dim: #6b551c;
    --radius: 10px;
  }

  *{margin:0;padding:0;box-sizing:border-box;}

  html{scroll-behavior:smooth;}
```

Browser Output — Playwright Screenshot `emergency-resource-map.html`



Output Verification: Floating search bar, category chips, and live facility listing with distance calculations.

4. Design Tokens & Color Palette Stylesheet

`static/css/tokens.css`

CSS Custom Properties declaring the design system: neutral surface values, category accents (hospital, shelter, water, food), elevation drop-shadows, and typography scales.

💻 **FULL SOURCE CODE: STATIC/CSS/TOKENS.CSS**

```
:root {
  /* Color Palette - Neutrals */
  --bg-app: #0f172a;
  --bg-surface: #ffffff;
  --bg-surface-elevated: #f8fafc;
  --bg-surface-subtle: #f1f5f9;
  --bg-overlay: rgba(15, 23, 42, 0.65);

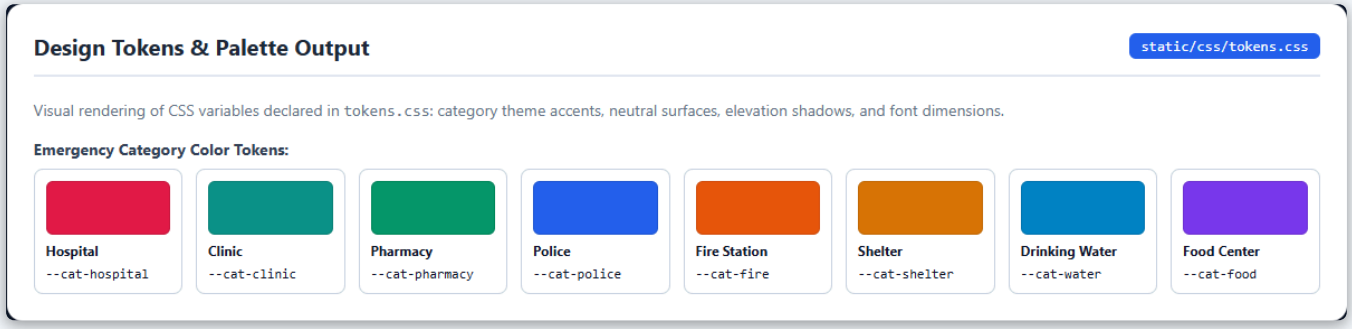
  --border-subtle: #e2e8f0;
  --border-strong: #cbd5e1;
  --border-focus: #2563eb;

  --text-primary: #0f172a;
  --text-secondary: #475569;
  --text-muted: #64748b;
  --text-inverse: #ffffff;

  /* Accent Colors */
  --color-primary: #2563eb;
  --color-primary-hover: #1d4ed8;
  --color-primary-light: #eff6ff;

  --color-danger: #dc2626;
  --color-danger-light: #fef2f2;
  --color-danger-border: #fecaca;
  --color-warning: #d97706;
  --color-warning-light: #fffbeb;
  --color-warning-border: #fde68a;
  --color-success: #16a34a;
  --color-success-light: #f0fdf4;
  --color-info: #0284c7;
```

🎨 **Rendered CSS Output Showcase: Design Tokens & Palette** — `tokens.css`



Output Verification: Rendered palette swatches showing `--cat-hospital`, `--cat-clinic`, `--cat-pharmacy`, and core brand tokens.

5. Layout Structure & Responsive Grid Stylesheet

`static/css/layout.css`

Styles governing viewport containment, fullscreen Leaflet map positioning, floating command island, top search bar wrapper, and responsive breakpoint adjustments.

💻 **SOURCE CODE (EXCERPT): STATIC/CSS/LAYOUT.CSS**

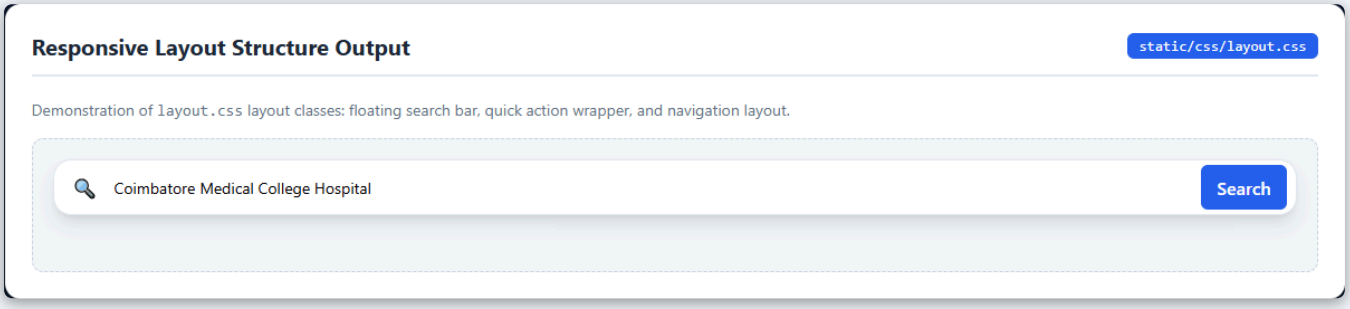
```
/* Base reset and layout */
*, *::before, *::after {
  box-sizing: border-box;
  margin: 0;
  padding: 0;
}

html, body {
  width: 100%;
  height: 100%;
  overflow: hidden;
  font-family: var(--font-sans);
  font-size: 16px;
  color: var(--text-primary);
  background-color: var(--bg-app);
  -webkit-font-smoothing: antialiased;
  -moz-osx-font-smoothing: grayscale;
}

button, input, select, textarea {
  font-family: inherit;
  font-size: inherit;
}

a {
  color: var(--color-primary);
  text-decoration: none;
}
a:hover {
  text-decoration: underline;
}
```

🎨 **Rendered CSS Output Showcase: Layout & Search Box Wrapper** `layout.css`



Output Verification: Rendered floating search wrapper with icon button, input box, and primary action trigger.

6. UI Components, Chips & Cards Stylesheet

`static/css/components.css`

Component styling rules for interactive category filter chips with colored indicator dots, emergency warning notice banners, facility cards, and action buttons.

💻 **SOURCE CODE (EXCERPT): STATIC/CSS/COMPONENTS.CSS**

```
/* Component Styles */

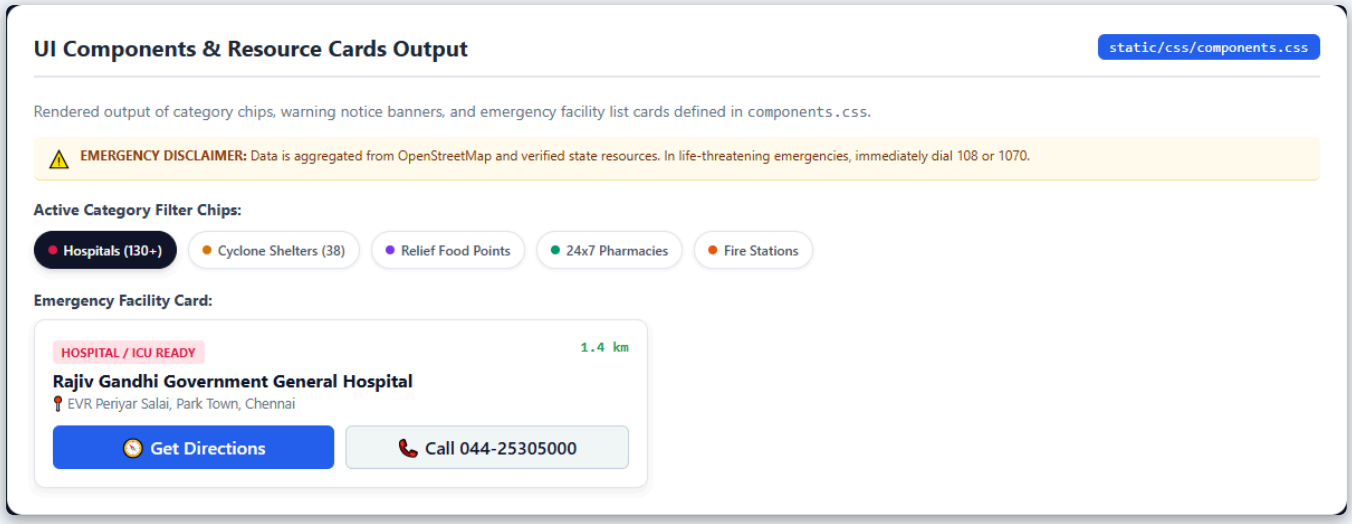
/* 1. Category Filter Chips */
.category-chip {
  display: inline-flex;
  align-items: center;
  gap: 6px;
  background: var(--bg-surface);
  border: 1px solid var(--border-subtle);
  border-radius: var(--radius-full);
  padding: 6px 12px;
  font-size: var(--text-xs);
  font-weight: 600;
  color: var(--text-secondary);
  cursor: pointer;
  white-space: nowrap;
  user-select: none;
  box-shadow: var(--shadow-sm);
  transition: all var(--transition-fast);
  min-height: 34px;
}

.category-chip:hover {
  border-color: var(--border-strong);
  color: var(--text-primary);
  background-color: var(--bg-surface-elevated);
}

.category-chip.active {
  background: var(--text-primary);
  color: var(--text-inverse);
```

Rendered CSS Output Showcase: Chips, Notice Banner & Facility Card

components.css



Output Verification: Rendered category chips, warning notice banner, and hospital facility card with action triggers.

7. Draggable Bottom Sheet & Detail Drawer Stylesheet `static/css/bottom-sheet.css`

CSS specifications for mobile draggable bottom sheets: handle bar styling, summary row, touch-action configuration, and expansion states (peek, half, full).

💻 **FULL SOURCE CODE: STATIC/CSS/BOTTOM-SHEET.CSS**

```
/* Mobile Draggable Bottom Sheet */

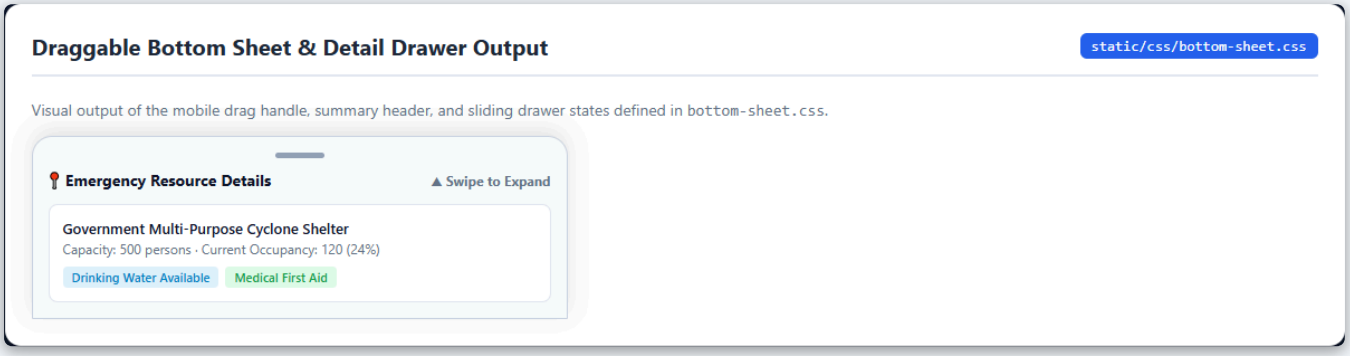
.bottom-sheet-container {
  display: none;
}

@media (max-width: 899px) {
  .bottom-sheet-container {
    display: flex;
    flex-direction: column;
    position: fixed;
    left: 0;
    right: 0;
    bottom: 0;
    height: 90vh;
    background: var(--bg-surface);
    border-top-left-radius: 20px;
    border-top-right-radius: 20px;
    box-shadow: 0 -8px 24px rgba(0, 0, 0, 0.2);
    z-index: var(--z-bottom-sheet);
    transform: translateY(calc(100% - 76px)); /* Peek height */
    transition: transform 0.28s cubic-bezier(0.2, 0.9, 0.3, 1);
    will-change: transform;
    overflow: hidden;
  }

  .bottom-sheet-container.dragging {
    transition: none; /* No transition while user fingers are dragging */
  }

  .bottom-sheet-container.state-peek {
```

🎨 Rendered CSS Output Showcase: Bottom Sheet Drawer `bottom-sheet.css`



Output Verification: Draggable pill handle bar, summary header, and expandable facility detail card.